

GATTON, Australia

WMO: 945620

Lat: 27.5436S

Lon: 152.3375E

Elev: 292

StdP: 14.54

Time Zone: 10.00 (E10)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	38.4	40.9	23.6	18.1	62.5	27.1	21.3	58.9	30.3	65.3	24.3	60.7	5.9	260	0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.1	97.1	71.2	93.8	71.1	90.9	70.4	76.0	88.9	74.7	86.6	73.6	84.7	10.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.6	122.3	80.2	71.5	117.6	78.9	70.5	113.5	78.3	39.6	89.5	38.4	86.8	37.4	85.2	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.1	18.4	16.1	32.0	104.3	2.1	4.2	30.5	107.3	29.3	109.7	28.2	112.1	26.7	115.1		
(5)			30.2	78.3	2.0	1.6	28.7	79.4	27.5	80.4	26.4	81.3	24.9	82.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	68.3	78.4	77.1	74.3	68.8	62.6	58.5	57.0	58.8	64.7	69.6	74.0	76.4	(6)
(7)		DBStd	8.61	4.08	4.21	3.73	3.53	3.76	3.82	3.81	3.98	4.67	4.96	5.03	4.81	(7)
(8)		HDD50	2	0	0	0	0	0	1	1	0	0	0	0	0	(8)
(9)		HDD65	823	0	0	0	7	93	197	249	200	62	14	1	0	(9)
(10)		CDD50	6677	879	759	752	563	392	255	219	272	440	607	720	819	(10)
(11)		CDD65	2024	414	339	287	121	19	2	2	6	51	156	272	355	(11)
(12)		CDH74	18367	3812	3016	2177	959	230	20	33	175	778	1558	2418	3191	(12)
(13)		CDH80	7168	1714	1265	712	187	17	1	1	35	245	578	1005	1408	(13)
(14)	Wind	WSAvg	7.1	7.4	7.3	7.0	6.2	6.7	7.1	7.3	7.5	7.2	7.2	7.4	7.2	(14)
(15)	Precipitation	PrecAvg	31.3	4.6	3.9	2.9	2.0	2.3	1.5	1.7	1.1	1.2	2.6	3.5	4.0	(15)
(16)		PrecMax	42.8	17.8	11.8	8.6	12.8	17.5	12.6	12.0	3.6	3.5	9.3	8.6	10.4	(16)
(17)		PrecMin	14.0	0.6	0.6	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.2	0.9	(17)
(18)		PrecStd	7.6	3.3	2.7	2.0	2.5	3.0	2.2	2.2	0.9	1.0	1.9	2.2	2.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	100.5	100.5	94.9	87.2	82.2	76.5	77.9	84.2	91.9	96.3	99.0	100.6	(19)
(20)			MCWB	73.7	74.6	72.3	67.6	63.3	61.2	59.8	61.3	63.8	65.7	68.7	70.9	(20)
(21)		2%	DB	95.7	94.3	89.7	84.3	78.9	73.4	73.8	78.5	86.8	90.3	93.8	95.4	(21)
(22)			MCWB	72.7	73.7	71.5	66.7	62.0	60.0	57.9	58.9	62.5	65.4	68.8	71.5	(22)
(23)		5%	DB	92.1	90.8	86.7	82.1	76.6	71.1	70.9	74.7	82.1	86.2	89.6	91.0	(23)
(24)			MCWB	72.0	72.2	70.0	66.0	61.0	58.6	57.2	57.2	61.3	65.0	68.3	71.3	(24)
(25)		10%	DB	88.8	87.4	84.0	79.6	74.1	68.7	68.2	71.7	78.2	82.6	86.0	87.5	(25)
(26)			MCWB	71.8	71.1	69.4	65.3	60.3	57.7	55.8	56.0	60.6	64.0	67.7	70.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.7	78.1	75.6	71.8	67.5	64.8	65.7	64.6	68.2	72.2	74.4	76.1	(27)
(28)			MCDWB	90.9	90.9	85.9	82.3	77.6	70.3	71.7	76.9	81.2	85.3	89.1	91.2	(28)
(29)		2%	WB	76.0	76.0	73.8	69.8	65.3	62.8	62.2	62.0	66.0	69.7	72.5	74.5	(29)
(30)			MCDWB	88.4	89.0	83.8	79.4	74.2	69.5	67.8	72.5	78.1	81.5	85.7	88.0	(30)
(31)		5%	WB	74.7	74.4	72.7	68.2	63.6	61.3	59.9	60.1	64.3	67.8	70.9	73.3	(31)
(32)			MCDWB	86.3	85.8	82.0	77.6	72.2	67.6	66.6	70.0	76.5	79.3	83.8	85.5	(32)
(33)		10%	WB	73.5	73.0	71.5	66.8	62.1	59.8	57.8	58.1	62.6	66.1	69.4	72.1	(33)
(34)			MCDWB	83.7	83.6	79.6	76.4	70.7	66.2	65.4	68.3	74.1	77.2	81.0	82.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	20.1	20.0	19.8	23.1	24.4	21.9	25.0	28.7	28.4	25.4	22.8	21.3	(35)
(36)			MCDBR	26.6	26.0	24.5	26.1	27.8	25.6	29.0	33.8	34.9	32.4	28.8	27.5	(36)
(37)			MCWBR	8.2	8.1	8.6	10.9	13.3	13.6	15.2	16.7	14.5	12.2	9.3	8.7	(37)
(38)		5% WB	MCDWB	20.5	22.4	19.4	22.4	22.3	19.4	22.5	27.8	27.3	25.0	23.6	22.0	(38)
(39)			MCWBR	7.3	7.7	7.4	10.5	11.5	11.3	13.4	15.4	12.4	10.2	8.6	7.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.386	0.373	0.353	0.333	0.312	0.304	0.295	0.308	0.349	0.373	0.382	0.394	(40)	
(41)		taud	2.481	2.526	2.576	2.610	2.615	2.635	2.644	2.586	2.467	2.433	2.451	2.442	(41)	
(42)		Ebn at Noon	304	302	299	290	283	279	287	295	295	298	303	302	(42)	
(43)		Edh at Noon	37	35	31	28	25	23	24	28	34	38	38	39	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2078	1901	1649	1443	1201	996	1147	1436	1754	1945	2106	2094	(44)	
(45)		RadStd	176	176	98	71	66	75	90	109	143	192	203	219	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		+0.74	+0.94	N/A	N/A	N/A	N/A	N/A	-121	+335	+202

Nomenclature: See separate page